



AI-Powered Fraud Detection and Prevention in Banking

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List of Abbreviations

AI - Artificial Intelligence

API - Application Programming Interface

CSV - Comma-Separated Values

DSR - Design Science Research

GDPR - General Data Protection Regulation

JSON - JavaScript Object Notation

KYC - Know Your Customer

ML - Machine Learning

PoC - Proof of Concept

PSD2 - Payment Services Directive 2

RF - Random Forest

SMOTE - Synthetic Minority Over-sampling Technique

USD - United States Dollar

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Financial fraud has scaled in parallel with the digital transformation of banking, exposing the limitations of rule-based detection systems that depend on fixed thresholds and manually maintained logic. Such systems produce high false-positive volumes and lag behind evolving fraud tactics, motivating a shift toward adaptive machine learning approaches. This thesis presents a proof-of-concept fraud detection system developed using the Design Science Research methodology. Three modules form the system: a configurable Python-based synthetic transaction generator using the Faker library; a Random Forest classification engine with one-hot encoding, temporal feature extraction, and class-imbalance handling via RandomOverSampler and a Streamlit-based dashboard enabling interactive exploration of detection outcomes. The generator produces 10,000 transactions with controllable fraud triggers covering high-risk geographies, off-hours timing, and high-value amounts, sidestepping the regulatory barriers to using real banking records for academic research. Stratified 70:30 splitting was used to preserve the fraud rate across training and test partitions, with RandomOverSampler applied only during training to prevent leakage into evaluation. The classifier reached 78.53% accuracy on unseen data, with fraud-class precision of 33.13%, recall of 34.10%, and an F1-score of 33.61% reflecting the limited minority-class signal available in a 10,000-record dataset. To address the regulatory demand for transparent decisions, each flagged transaction is paired with a human-readable fraud reason label produced by a rule-based annotator, complementing the model's prediction with an interpretable justification. The thesis closes by identifying limitations and future directions, including real-time streaming integration, multi-algorithm comparison, larger datasets and cloud deployment for production-grade workloads.
Key words fraud detection, machine learning, random forest classifier, class imbalance handling, synthetic data generation

1 Introduction

The banking and financial sector has undergone a profound digital transformation over the past two decades. This transformation has brought unprecedented convenience to consumers through mobile applications, digital wallets, contactless payments, and instant online transfers. However, this digitalization has simultaneously created new vulnerabilities and expanded the attack surface available to fraudsters. Modern financial fraud encompasses a wide spectrum of criminal activities including unauthorized transactions, identity theft, synthetic identity fraud, account takeover attacks, and sophisticated social engineering scams that exploit both technological vulnerabilities and human psychology.

The scale and impact of financial fraud are substantial. According to industry reports, global losses attributed to financial fraud amount to billions of dollars annually, placing immense operational and reputational pressure on financial institutions worldwide. Traditional fraud detection systems have predominantly relied on rule-based engines that flag transactions based on predefined conditions and static thresholds. While these systems remain widely deployed, they suffer from fundamental limitations including rigidity, high false positive rates, and vulnerability to adversaries who understand and circumvent the established rules.

The regulatory environment has further intensified the demands placed on fraud detection systems. Frameworks such as the Payment Services Directive 2 in Europe and Know Your Customer requirements globally mandate not only effective fraud detection but also transparent, accountable, and non-discriminatory decision-making processes. Financial institutions must demonstrate the rationale behind their fraud detection decisions, maintain comprehensive audit trails, and ensure that their systems do not introduce undue bias against particular demographic groups.

These converging pressures have motivated a paradigm shift toward intelligent, adaptive systems that leverage artificial intelligence and machine learning. Machine learning algorithms offer several advantages over traditional methods: they can learn complex patterns from historical data, adapt to new fraud strategies through periodic retraining, handle high-dimensional feature spaces, and provide probabilistic risk scores rather than binary rule-based decisions. This thesis explores the practical application of machine learning algorithm in developing a fraud detection proof of concept for the banking sector.

1.1 Background and Motivation

Traditional rule-based fraud detection systems operate by applying expert-defined rules and static thresholds to incoming transactions. A transaction might be flagged if it exceeds a monetary threshold, originates from a high-risk geographic location, or occurs at an unusual time. While these systems are straightforward to implement and effective for known fraud patterns, they cannot adapt to evolving fraud tactics. The manual process of updating rules creates an inherently reactive approach where institutions continually attempt to catch up with new fraud methodologies rather than proactively anticipating them.

Machine learning algorithms address these limitations by learning patterns directly from historical transaction data. They can identify complex, non-linear relationships between features that would be difficult or impossible to encode as explicit rules. Furthermore, ML models can be periodically retrained on new data, enabling them to adapt to changing fraud patterns without manual rule updates. This adaptive capability is particularly important in the context of financial fraud, where adversaries continuously evolve their tactics to evade detection.

The motivation for this thesis stems from the recognition that while machine learning has shown promising results in fraud detection research, there remains a need for comprehensive, practical demonstrations of end-to-end fraud detection systems. Sanusi et al. (2015) provide an overview of the principal fraud schemes affecting banking institutions and the preventive measures used to limit financial loss, situating the technical detection problem within its broader operational context. Much of the existing literature focuses on algorithmic comparisons using established benchmark datasets, but there is limited guidance on how to construct a complete fraud detection pipeline including data generation, preprocessing, model training, and operational visualization. This thesis addresses this gap by implementing and evaluating a complete proof-of-concept system.

1.2 Objectives and Research Questions

The primary objective of this thesis is to design, implement, and evaluate a proof-of-concept system for AI-powered fraud detection in banking using a design science research methodology. The specific objectives are:

- To identify the key functional and non-functional requirements for an AI-powered fraud detection system in the banking sector through a comprehensive literature review

- To generate realistic synthetic transaction data that incorporates configurable fraud indicators, addressing the challenge of real data unavailability for research purposes
- To apply supervised machine learning, specifically the Random Forest algorithm, for fraud classification with appropriate preprocessing and class imbalance handling techniques
- To develop an interactive dashboard for real-time visualization, analysis, and export of fraud detection results
- To evaluate system performance using standard classification metrics and assess the practical utility of the system through qualitative analysis

The following research questions guide the investigation:

RQ1: What are the key functional and non-functional requirements for developing an AI-powered fraud detection system in the banking sector?

RQ2: How can machine learning algorithms be effectively utilized to detect fraudulent activities in banking transactions?

RQ3: What are the major challenges and limitations in implementing an AI-based fraud detection system, and how can they be addressed?

RQ4: How does the proof-of-concept demonstrate the feasibility and effectiveness of AI technologies in enhancing fraud detection capabilities compared to traditional methods?

1.3 Scope and Delimitations

This thesis is delimited to requirement analysis and proof-of-concept development. It employs synthetic data rather than real banking data due to privacy and regulatory constraints. The machine learning implementation focuses on the Random Forest algorithm. A comprehensive multi-algorithm comparison involving deep learning and gradient boosting methods is excluded from the current scope but identified as a direction for future work.

The system processes data in batch mode rather than truly streaming real-time data, although the architecture is designed to accommodate future streaming integration. Full-scale production deployment, infrastructure cost analysis, and integration with live banking APIs are beyond the scope of this work. The evaluation focuses on technical performance metrics and qualitative assessment of system utility rather than field testing with real users or institutions.

1.4 Structure of the Thesis

This thesis is organized into seven chapters. Chapter 1 introduces the research context, objectives and research questions. Chapter 2 establishes the theoretical framework through a review of relevant literature on fraud detection methods, machine learning approaches, and related challenges. Chapter 3 describes the research methodology, including the design science research approach, data collection and analysis methods, and the evaluation framework.

Chapter 4 presents the system design and implementation in detail, covering data generation, preprocessing, the machine learning model, and the interactive dashboard. Chapter 5 reports the results and evaluation of the system. Chapter 6 discusses the findings in relation to the research questions and the broader literature. Chapter 7 concludes the thesis with a summary, key contributions, and recommendations for future work. The appendices provide supplementary technical documentation and screenshots of the system.

Use of generative artificial intelligence - in line with Haaga-Helia's Guidelines for the use of Artificial Intelligence, the author acknowledges the use of Anthropic's Claude during the preparation of this thesis. Claude was used as an assistant for proofreading and language checking of author-written text, for structural feedback on chapter organisation, and for formatting the document to comply with Haaga-Helia's reporting guidelines. The research problem, objectives and research questions, literature review and selection of sources, methodology, the Python implementation of the synthetic data generator, the machine learning detection engine and the Streamlit dashboard, as well as the experimental design, model evaluation, and all findings, analyses, and conclusions were authored independently by the author without AI generation. No part of the final thesis text was AI-generated, and AI tools were not used as a source of factual or scholarly content. All AI-assisted edits were reviewed and verified by the author, who bears sole responsibility for the final work.

2 Theoretical Framework

Understanding the theoretical foundations of fraud detection requires engagement with multiple interdisciplinary domains, including financial services, machine learning, regulatory compliance, and behavioral analytics. This chapter synthesizes contemporary knowledge across these domains to establish the conceptual and technical foundations supporting the development of intelligent fraud detection systems. The framework presented below integrates traditional approaches that have shaped fraud detection practices with cutting-edge machine learning methodologies that represent the current state of the art. Additionally, this chapter addresses critical contemporary challenges including class imbalance in fraud datasets, regulatory compliance requirements, and the imperative for explainability in algorithmic decision-making. By establishing this comprehensive theoretical foundation, the research grounds its practical contributions within established academic discourse and contemporary industry practice.

2.1 Traditional Fraud Detection Methods

Before the emergence of machine learning approaches, fraud detection in banking relied primarily on rule-based systems and statistical anomaly detection techniques. These traditional methods continue to form the operational backbone of many contemporary fraud detection systems, though increasingly supplemented by machine learning components. Understanding traditional approaches provides important context for appreciating the limitations that motivate machine learning adoption and the evolutionary development of fraud detection capabilities.

2.1.1 Rule-Based Systems

The earliest and most widespread fraud detection systems in the banking sector are rule-based. These systems operate by applying expert-defined rules and static thresholds to incoming transactions, flagging those that violate predefined conditions as potentially fraudulent. For instance, a transaction might be flagged if it exceeds a certain monetary threshold, originates from a geographic location known for high fraud incidence, or occurs at an unusual time of day relative to the customer profile.

The principal advantages of rule-based systems include their ease of implementation, interpretability, and suitability for detecting well-known fraud patterns. However, they suffer from several significant limitations: high false positive rates that burden investigation teams,

inflexibility in adapting to novel or evolving fraud tactics, and vulnerability to adversaries who understand and deliberately circumvent the defined rules. Despite these shortcomings, rule-based systems remain a foundational component in many multi-layered fraud detection architectures.

2.1.2 Statistical and Anomaly-Based Systems

As fraud techniques grew in sophistication, financial institutions began implementing statistical and anomaly-based detection systems. Bolton and Hand (2002) provide an early comprehensive review of these statistical approaches, identifying challenges that persist in contemporary anomaly-based fraud detection. These systems establish a baseline profile of normal transaction behaviour and flag deviations as potential anomalies. Common techniques include statistical profiling, time-series analysis, and behavioural scoring.

The key advantage of these approaches is that they do not require labeled fraud data and can therefore detect previously unseen fraud patterns. However, calibrating the sensitivity of anomaly detection is challenging settings that are too strict generate excessive false positives, while lenient settings allow actual fraud to pass undetected. Furthermore, these systems require large volumes of historical data to establish reliable baselines and often lack the contextual understanding necessary to distinguish between genuine anomalies and benign behavioral changes. More recent reviews of anomaly detection in financial fraud, such as Hilal et al. (2022), confirm that these limitations have continued to motivate the move toward learning-based approaches.

2.1.3 Commercial Fraud Detection Platforms

Several proprietary platforms are widely adopted in the financial sector, offering scalable, high-throughput systems with integrated machine learning capabilities and visualization tools for analysts.

Table 1. Commercial Fraud Detection Platforms

Platform	Vendor	Key Capabilities
FICO Falcon Fraud Manager	FICO	AI/ML models, real-time scoring, deployed at 10,000+ global financial institutions

		(FICO 2026).
SAS Detection and Investigation for Banking	SAS	Hybrid analytics (rules + ML + network analysis), behavior profiling, real-time monitoring
IBM Safer Payments	IBM	User-configurable ML models, rules + ML hybrid detection, high-throughput real-time processing
Oracle FCCM	Oracle	Rule-based behavior detection, graph/network analysis, integrated AML and compliance reporting

2.2 Machine Learning in Fraud Detection

Contemporary machine learning techniques have transformed fraud detection by enabling systems to identify complex, nonlinear patterns in transaction data that elude traditional statistical methods. The systematic review by Ali et al. (2022) confirms this shift, finding that ensemble and deep-learning methods consistently outperform rule-based baselines across recent banking-fraud studies. The application of data mining techniques in financial fraud detection has earlier roots - Ngai et al. (2011) provide a foundational classification framework that organised the field's evolution from rule engines to data-mining approaches and remains widely cited. Machine learning approaches learn fraud indicators directly from historical data rather than relying on manually specified rules, creating more flexible, adaptive detection capabilities. This section surveys the principal machine learning methodologies applied to fraud detection contexts, with particular emphasis on ensemble methods and classification algorithms relevant to banking applications.

2.2.1 Supervised Learning

Supervised machine learning approaches require labeled historical data containing examples of both fraudulent and legitimate transactions. Classification algorithms learn decision boundaries that separate fraud from non-fraud based on transaction features. Logistic regression represents a foundational supervised approach, modeling the probability of fraud as a function of transaction features using a sigmoid decision boundary. While logistic

regression offers interpretability advantages and establishes useful baseline performance, it assumes linear relationships between features and fraud probability, limiting its capacity to capture complex fraud patterns. Comparative studies such as Vorobyev and Krivitskaya (2022) report that ensemble classifiers consistently outperform single-model alternatives across diverse credit card fraud detection benchmarks.

Decision trees provide alternative classification approaches, recursively partitioning the feature space to create hierarchical decision rules. Individual decision trees suffer from high variance and overfitting tendencies, particularly on complex fraud detection tasks with high-dimensional feature spaces.

Random Forest algorithms address these limitations by aggregating predictions across numerous randomly constructed decision trees, substantially improving generalization performance and robustness to overfitting.

2.2.2 Random Forest Classification

Random Forest algorithms represent ensemble methods that construct multiple decision trees during training and aggregate their predictions during inference. Each tree in the ensemble is trained on a random bootstrap sample of the training data, and at each node split, the algorithm considers a random subset of features, introducing both data level and feature-level randomness. During inference, classification decisions result from aggregating predictions across all trees, typically through majority voting for classification tasks.

Random Forest methods demonstrate several characteristics making them particularly suitable for fraud detection applications. They naturally handle nonlinear relationships and complex feature interactions without explicit feature engineering. They provide built in feature importance estimates, useful for understanding which transaction attributes most strongly predict fraud. They achieve strong performance on imbalanced datasets, a critical consideration in fraud detection where fraudulent transactions typically comprise less than 1% of volumes. They are relatively robust to hyperparameter choices and scale effectively to large datasets. Breiman (2001) established the theoretical properties of Random Forests and demonstrated their practical effectiveness across diverse classification domains.

2.2.3 Gradient Boosting Methods

Gradient boosting represents an alternative ensemble approach that sequentially constructs trees, with each successive tree designed to correct errors made by previous ensemble members. XGBoost, LightGBM, and CatBoost implementations of gradient boosting have achieved state-of-the-art performance on numerous machine learning competitions and production systems.

Gradient boosting methods offer advantages including superior predictive performance on many tasks, more sophisticated handling of missing data, and built-in regularization mechanisms. However, gradient boosting introduces greater hyperparameter complexity and typically requires more careful tuning compared to Random Forest methods. For fraud detection contexts specifically, gradient boosting approaches require careful management of class imbalance through techniques such as `scale_pos_weight` adjustment and threshold optimization.

Table 2. Machine Learning Approaches in Fraud Detection

Algorithm	Description	Advantages	Limitations
Logistic Regression	Binary classification with interpretable coefficients	Fast, interpretable	Linear decision boundary
Decision Trees	Rule-based hierarchical classification	Interpretable, handles non-linear patterns	Prone to overfitting
Random Forest	Ensemble of decision trees	Robust, feature importance, handles mixed types	Less interpretable than single tree
Gradient Boosting	Sequential ensemble learning	High accuracy	Computationally intensive
Neural Networks	Multi-layer perceptron	Complex pattern learning	Requires large data, black box

2.3 Class Imbalance in Fraud Detection

Class imbalance is a fundamental challenge in fraud detection where fraudulent transactions constitute a small minority of total transactions, typically less than 1-2%. Standard machine learning algorithms trained on imbalanced datasets tend to be biased toward the majority class, achieving high overall accuracy while performing poorly on the minority fraud class.

Several techniques have been developed to address class imbalance. Data-level approaches such as Random Oversampling, SMOTE (Synthetic Minority Over-sampling Technique), and Random Undersampling modify the training dataset to balance class distributions. Algorithm-level approaches such as cost-sensitive learning and threshold adjustment modify the learning algorithm itself to pay greater attention to the minority class. Ensemble methods combine multiple models trained on different data samples or with different parameters to improve minority class detection.

Research has demonstrated that combining resampling techniques with appropriate algorithm configurations can significantly improve fraud detection performance on imbalanced datasets. This thesis employs RandomOverSampler applied exclusively to training data, avoiding test set contamination while addressing class imbalance.

2.4 Regulatory Landscape in Financial Services

Contemporary fraud detection systems must operate within complex, evolving regulatory frameworks that impose specific requirements on detection methodologies, data handling, and decision documentation. Understanding these regulatory requirements is essential for developing compliant systems, as regulatory violations carry severe penalties and reputational consequences for financial institutions.

2.4.1 General Data Protection Regulation (GDPR)

The European Union's General Data Protection Regulation, effective since May 2018, establishes stringent requirements for handling personal data and makes specific provisions relevant to algorithmic decision-making. GDPR Article 22 restricts automated decision-making that produces legal or similarly significant effects, requiring human involvement and providing individuals rights to request explanations. Fraud detection systems, particularly those making decisions affecting customer account access or transaction blocking, fall within this provision's scope. Compliance requires that fraud detection systems incorporate human

review processes, maintain audit trails documenting decision rationales, and provide customers with explanations when automated systems deny transactions or block accounts. GDPR additionally mandates data minimization, requiring that fraud detection systems collect only data necessary for their specific purpose, and establishes retention limits preventing indefinite storage of customer transaction records. The right to be forgotten creates additional complexities for fraud detection, as customers may request deletion of their personal data, potentially conflicting with the retention of historical transaction patterns necessary for fraud model training and maintenance.

2.4.2 Payment Services Directive 2 (PSD2)

PSD2, implemented across European Union member states, establishes technical and organizational requirements for payment service providers and strong customer authentication mechanisms. The directive requires that payment service providers implement strong customer authentication combining at least two independent factors from the categories of knowledge, possession, and inherence. For fraud detection specifically, PSD2 recognizes risk-based authentication approaches that may exempt certain low-risk transactions from full authentication requirements, provided the payment service provider implements appropriate fraud detection and monitoring capabilities. PSD2 further establishes requirements for transaction monitoring, fraud detection capabilities, and reporting obligations to financial intelligence units when suspicious activities are detected. The directive's Open Banking provisions enable third-party access to payment data, creating both opportunities and challenges for fraud detection systems that must operate in contexts where transaction data flows across multiple institutional boundaries.

2.4.3 Anti-Money Laundering (AML) Directives

Financial institutions face legal obligations under Anti-Money Laundering directives to implement systems detecting suspicious transaction patterns potentially indicative of money laundering, terrorist financing, or other financial crimes. AML obligations require customer due diligence establishing customer identity and beneficial ownership, transaction monitoring identifying suspicious patterns, and reporting suspicious activities to financial intelligence units within specified timeframes. While distinct from fraud detection, AML requirements create overlapping concerns with fraud detection systems regarding transaction monitoring, feature engineering, and alert management. AML systems focus on patterns suggesting criminal proceeds laundering such as rapid funds movements across multiple institutions, structuring behavior splitting deposits below reporting thresholds, or geographic

inconsistencies with legitimate business operations rather than transaction-specific fraud indicators. Contemporary institutional practice increasingly integrates fraud and AML monitoring capabilities within unified risk analytics platforms, creating synergies in data infrastructure, feature engineering, and alert investigation workflows.

2.5 Explainable Artificial Intelligence in Financial Services

The increasing deployment of machine learning systems in financial decision-making has generated substantial emphasis on explainability, interpretability, and transparency. Regulators, customers, and fraud analysts all require understanding of how algorithmic systems reach decisions, particularly when those decisions affect customer rights or impose restrictions on financial access. Explainable AI represents a diverse set of techniques enabling humans to understand machine learning model predictions, moving beyond black-box models that provide classification decisions without interpretable explanations.

2.5.1 Explainability Versus Accuracy Trade-offs

A fundamental tension exists in machine learning between predictive accuracy and model interpretability. Linear models such as logistic regression offer straightforward interpretability - feature coefficients directly indicate their contribution to predictions but often sacrifice accuracy on complex nonlinear problems. Black-box models like neural networks and gradient boosting often achieve superior accuracy but provide little interpretability regarding their decision processes. Fraud detection presents a context where this trade-off carries acute significance. Fraud analysts, who typically lack machine learning expertise, require interpretable explanations to validate model decisions and identify potential system errors. Conversely, high detection accuracy is critical for operational effectiveness. Explainable AI techniques enable navigating this trade-off by overlaying interpretability onto inherently high-accuracy models.

2.5.2 SHAP (SHapley Additive exPlanations) Values

SHAP values represent a principled approach to model explainability grounded in cooperative game theory, specifically Shapley values from economics. SHAP values quantify each feature's contribution to a model's prediction for a specific instance, answering the question: 'How much did this feature contribute to moving this prediction away from the model's average prediction?' SHAP values satisfy important theoretical properties including local accuracy (explanations sum to the difference between the instance's prediction and the average), consistency (features with greater impact in one model than another receive higher

SHAP values in the first model), and symmetry (feature importance order is invariant to feature ordering in the model). For fraud detection, SHAP values enable fraud analysts to understand precisely which transaction features triggered fraud alerts. A SHAP analysis might reveal that a transaction's high-risk country combination contributed most heavily to a fraud flagging, followed by the late-night timing of the transaction, providing actionable insights for review analysts. SHAP values support both local explanations for individual predictions and global explanations characterizing how features typically influence model behavior.

2.5.3 Feature Importance and Model Transparency

Beyond individual prediction explanations, understanding which features the model considers most important for fraud detection supports both system validation and regulatory compliance. Random Forest algorithms naturally provide feature importance estimates based on the aggregate contribution of each feature across all trees, measured through techniques such as Gini importance or permutation importance. Feature importance analysis serves multiple purposes in fraud detection contexts. It provides model validation, ensuring that detected important features align with domain knowledge about fraud patterns. Unexpected feature importance can signal data quality issues or erroneous training data that requires investigation. Feature importance guides feature engineering efforts, identifying which transformations and engineered features most effectively contribute to model performance. For regulatory compliance, feature importance documentation supports explanations to regulators regarding how fraud detection systems operate.

2.6 Feature Engineering for Fraud Detection

Feature engineering - the process of constructing informative features from raw data represents a critical determinant of fraud detection system performance. Raw banking transaction data includes basic attributes such as transaction amounts, timestamps, merchant categories, and customer identifiers. Sophisticated feature engineering transforms these raw inputs into features that directly capture fraud indicators and customer behavior patterns. Feature engineering draws domain knowledge from fraud prevention specialists, financial analysts, and operational experience, combining this expertise with data-driven analysis to identify effective fraud predictors. Recent work such as Shabbir et al. (2022) has proposed multi-faceted feature extraction approaches that combine transaction attributes with behavioural indicators, illustrating how feature engineering continues to evolve in fraud detection contexts.

2.6.1 Transaction-Level Features

Transaction-level features describe properties of individual transactions independent of customer history. Amount-based features capture transaction magnitude, recognizing that fraud often exhibits characteristic amount distributions differing from legitimate transactions. Features such as log-transformed transaction amounts, deviation of transaction amount from customer's average transaction amount, and percentile rankings of transaction amounts relative to customer history encode fraud-relevant information. Temporal features capture time-of-day and day-of-week patterns, recognizing that legitimate transaction timing varies systematically by transaction type and customer behavior, while fraud often exhibits characteristic temporal patterns. Geographic features identify transaction location, comparing against customer's typical transaction geography and detecting transactions from implausible locations. Merchant features capture properties of the merchant accepting the transaction, including merchant category codes, merchant reputation scores, and historical fraud rates associated with specific merchants or merchant categories.

2.6.2 Customer Behavior Features

Customer-level features aggregate transaction properties to characterize customer behavior patterns and identify deviations indicative of account compromise or fraudulent activity. Historical features aggregate customer's previous transactions, computing statistics such as average transaction amount, transaction frequency, standard deviation of amounts, and count of transactions in various categories over preceding time windows. Behavioral deviation features quantify how a current transaction deviates from the customer's established patterns, computing z-scores of amount relative to customer's historical mean and standard deviation, deviation metrics for geographic and temporal patterns, and count of transactions deviating beyond specified thresholds. Velocity features capture transaction frequency and volume changes, identifying sudden spikes in activity that may indicate account compromise. Risk scoring features reflect accumulated customer risk based on previous fraudulent transactions, account age, customer verification completeness, and historical compliance issues.

2.6.3 Engineered Statistical Features

Beyond transaction-level and customer-level features, domain expertise enables construction of features capturing sophisticated fraud patterns. Traveling distance features compare geographic distance between consecutive transactions and time available for travel,

identifying implausible combinations of locations and timestamps suggesting geographic inconsistency. Multi-step fraud sequences identify patterns of transactions across multiple accounts or merchants within brief time windows, potentially indicating money laundering or account takeover cascades. Cross-customer patterns identify unusual concentrations of transactions between specific customer pairs or involvement in transaction networks exhibiting suspicious topological properties. Behavioral shift detection identifies statistically significant changes in customer behavior, flagging accounts suddenly exhibiting transaction patterns radically different from historical baselines. These sophisticated features require careful construction and validation to ensure they capture meaningful patterns rather than statistical artifacts.

2.7 Research Gap and Contribution

While substantial academic literature addresses fraud detection and machine learning applications in financial services, significant gaps remain regarding integrated treatment of accuracy optimization, real-time operational constraints, regulatory compliance and explainability requirements. Prior research frequently treats these dimensions separately or focuses on particular datasets and methodologies without sufficient generalization discussion. Additionally, practical implementation considerations such as data pipeline architecture, model deployment strategies, and fraud analyst integration receive limited treatment in academic literature despite their critical importance for operational effectiveness. This research contributes by presenting an integrated approach addressing the complete development lifecycle from problem definition through prototype implementation, evaluation, and deployment considerations. The specific focus on combining Random Forest classification, comprehensive feature engineering, transparent rule-based fraud reason labelling for explainability, and regulatory compliance requirements addresses a gap in literature integrating these multiple dimensions. The development and evaluation of a functional prototype supporting fraud analyst workflows bridges the gap between academic research and operational implementation, providing practical insights into translating fraud detection theories into production systems.

In summary, the theoretical framework presented in this chapter establishes the interdisciplinary foundations necessary for developing an effective fraud detection system. The evolution from rule-based systems to machine learning approaches demonstrates the increasing sophistication required in fraud detection, while the examination of ensemble methods particularly Random Forest algorithms - provides the methodological basis for the classification approach adopted in this research. The analysis of class imbalance challenges

informs the evaluation methodology and model training strategies, ensuring that performance assessment reflects operational reality rather than misleading accuracy metrics. The regulatory landscape analysis, encompassing GDPR, PSD2, and AML requirements, establishes binding constraints that the system architecture must satisfy, while the explainable AI literature provides techniques for meeting transparency requirements without sacrificing predictive performance. Finally, the feature engineering framework translates domain expertise into computational features that enable effective fraud discrimination. Together, these theoretical perspectives form a cohesive foundation supporting the integrated system development approach adopted in the subsequent chapters.

3 Research Methodology

This chapter describes the research methodology adopted for this investigation, encompassing the overall research paradigm, data collection and generation approaches, system development methodologies, and evaluation frameworks. The research adopts a design science research (DSR) approach, recognizing that the primary research objective involves developing an artifact - a functional fraud detection system and evaluating its effectiveness for addressing the identified problem. This methodology proves particularly appropriate for applied research contexts where practical system development constitutes the core research contribution, as opposed to pure empirical investigation of naturally occurring phenomena.

3.1 Research Approach: Design Science Research

This thesis follows the Design Science Research methodology, which is particularly suited for information systems research involving the creation and evaluation of artifacts. Design science research aims to create innovative artifacts that solve identified problems and contribute to both theoretical knowledge and practical application.

The DSR framework employed in this thesis follows the process model proposed by Peffers et al. (2007), which consists of six activities:

- Problem identification and motivation: Identifying the limitations of traditional fraud detection systems and the need for adaptive, AI-based approaches
- Objectives of a solution: Defining the requirements for an effective AI-powered fraud detection system
- Design and development: Implementing the proof-of-concept system including data generation, ML model, and dashboard
- Demonstration: Operating the system with synthetic data to demonstrate its capabilities
- Evaluation: Assessing system performance using classification metrics and qualitative analysis
- Communication: Documenting the research through this thesis and presenting findings

Table 3. DSR Activity Application in This Thesis

DSR Activity	Application in This Thesis
Problem Identification	Literature review on fraud detection limitations and regulatory requirements
Objectives Definition	Functional and non-functional requirements specification
Design and Development	Implementation of synthetic data generator, Random Forest classifier and Streamlit dashboard
Demonstration	System operation with configurable synthetic datasets of varying sizes
Evaluation	Performance metrics (accuracy, precision, recall, F1-score), confusion matrix analysis, feature importance
Communication	This thesis document and system documentation

3.2 Data Collection and Generation

Authentic fraud detection requires datasets containing both legitimate and fraudulent transaction examples with ground truth labels indicating actual transaction outcomes. However, access to real financial transaction data is restricted due to regulatory requirements and institutional confidentiality. This research employs synthetic data generation, creating realistic transaction data incorporating known fraud patterns and legitimate transaction behaviors without exposing proprietary banking data.

3.2.1 Synthetic Transaction Dataset

The research generated a synthetic dataset comprising 10,000 banking transactions, with fraud labels assigned probabilistically based on configurable risk factors. The synthetic data generation process incorporated domain knowledge about fraud patterns and legitimate transaction behaviors. The dataset includes transaction attributes encompassing amount, currency, timestamp, transaction type, device type, and geographic information. Feature

values were generated to reflect realistic distributions observed in financial data for example, transaction amounts following log-normal distributions reflecting the reality that very-high-value transactions are substantially less frequent than moderate-value transactions. Fraudulent transaction characteristics incorporated known fraud indicators: unusual amounts relative to customer's historical patterns, geographic implausibilities suggesting account compromise, late-night and weekend concentrations, and patterns involving high-risk merchant categories. Legitimate transactions were generated to exhibit realistic customer behavior regular salary deposits on consistent schedules, recurring utility payments, retail purchases with realistic amount and temporal distributions - creating datasets reflecting genuine banking operation patterns.

3.2.2 Data Characteristics and Validation

The 10,000-transaction dataset contains approximately 8,300 legitimate transactions and 1,700 fraudulent transactions, reflecting a synthetic fraud rate of approximately 17%. This rate, higher than typical real-world banking fraud prevalence of around 1%, results from the probabilistic fraud injection logic that independently evaluates each transaction against three configurable triggers (high-risk location, outside-hours timing, and high-value amount). The elevated rate provides sufficient minority-class examples for meaningful model training on a 10,000-record dataset. Approximately 70% of the data (7,000 transactions) was allocated to model training, with the remaining 30% (3,000 transactions) retained for final evaluation. Stratified sampling was used to preserve the fraud rate in both partitions. The implications of this elevated fraud representation for performance metric interpretation are discussed in Chapter 5.

3.3 System Analysis and Design

System design proceeded through structured analysis of fraud detection requirements, architectural pattern selection, and component specification. The fraud detection system architecture encompasses three primary layers: data ingestion and preprocessing, machine learning model inference, and fraud analyst interface.

3.3.1 Data Preprocessing Pipeline

Raw transaction data requires preprocessing before model inference, primarily to extract temporal information from timestamps and to convert categorical variables into a numerical representation suitable for the classifier. The preprocessing pipeline performs two main transformations. First, temporal feature extraction parses each Timestamp string to derive an

integer hour-of-day value and a binary `is_night` indicator (set when the hour falls outside normal business hours). Second, categorical features (`Transaction_Type`, `Location`, and `Device`) are encoded through one-hot encoding using pandas' `get_dummies` function, creating binary indicator columns for each category value. Because the dataset is generated synthetically with controlled distributions, no missing-value imputation or outlier handling is required in this proof-of-concept; both would be necessary additions for a production system operating on real banking data. The implementation details of these preprocessing steps are presented in Section 4.3.

3.3.2 Machine Learning Model Architecture

The core fraud detection model implements Random Forest classification using scikit-learn's `RandomForestClassifier`. The model was configured with 100 trees (`n_estimators=100`), balanced class weights (`class_weight='balanced'`), a fixed random seed for reproducibility (`random_state=42`), and parallel processing across all available CPU cores (`n_jobs=-1`). The `class_weight='balanced'` setting tells the model to pay more attention to fraud cases than to normal ones, since fraud is the minority class. This works alongside the oversampling step, which duplicates fraud rows to balance the training data together, they help the model take fraud seriously rather than ignoring it. Other tree-shape settings, such as how deep each tree can grow, were left at scikit-learn's default values. This means each individual tree is allowed to grow fully, but because predictions are averaged across 100 trees, the risk of overfitting is reduced. Tuning these settings more carefully is suggested as future work in Section 7.2.

3.4 Data Management and Privacy

Data management practices implemented throughout the research adhere to principles of data minimization, access control, and privacy preservation. Despite the use of synthetic rather than real customer data, appropriate privacy practices were maintained to establish design patterns suitable for deployment in production banking contexts. The research maintained clear separation between development, validation, and evaluation data, preventing data leakage where information from evaluation data influences model development. Transaction identifiers and customer identifiers, while necessary for system operation, were managed separately from feature data, limiting exposure of personal identifying information. Access controls in version management systems restricted access to code and data artifacts to authorized research personnel. Data retention policies specified

deletion of evaluation data upon research completion, consistent with GDPR principles of data minimization and limited retention.

3.5 Evaluation Framework

Evaluation of the fraud detection system employed multiple complementary metrics capturing different dimensions of system performance relevant to operational fraud detection.

3.5.1 Classification Metrics

Classification performance is evaluated using standard machine learning metrics derived from confusion matrix analysis. Precision measures the proportion of transactions flagged as fraudulent that actually represent fraud, directly addressing the false positive problem affecting fraud analyst efficiency. Recall measures the proportion of actual fraudulent transactions successfully detected, addressing detection completeness. F1-score computes the harmonic mean of precision and recall, providing a balanced metric when both metrics merit consideration. Receiver Operating Characteristic (ROC) curves characterize model performance across different decision thresholds, plotting true positive rate against false positive rate. Area Under the ROC Curve (AUC) provides a scalar summary of classification performance independent of decision threshold selection. Precision-Recall curves provide complementary threshold analysis, relevant for imbalanced fraud datasets where ROC curves may be misleading.

3.5.2 Business-Relevant Metrics

Beyond standard classification metrics, evaluation incorporated metrics directly relevant to fraud detection operations. Detection rate measures the percentage of fraudulent transactions successfully prevented through system alerts. Cost-benefit analysis evaluates the economic implications of different detection thresholds, weighting costs of false positives (legitimate transactions incorrectly blocked, creating customer friction) against costs of false negatives (fraudulent transactions escaping detection). Alert volume measures the total number of transactions requiring human analyst review, directly affecting operational staffing and efficiency. The research evaluated the impact of different decision thresholds on these business metrics, recognizing that optimal threshold selection depends on institutional preferences regarding precision-recall tradeoffs and acceptable alert volumes.

3.6 Prototype Development and Demonstration

The research developed a functional prototype incorporating the trained fraud detection model, transaction processing pipeline and interactive dashboard supporting fraud analyst workflows. The prototype demonstrates practical system operationalisation, moving beyond model development to system engineering required for production deployment. The dashboard provides fraud analysts with transaction-level details, predicted fraud labels, rule-based fraud reason annotations explaining why each transaction was flagged, model performance metrics, and interactive distribution-analysis views supporting fraud investigation workflows. The prototype is implemented as a Python-based Streamlit application with a modular architecture, supporting straightforward future extension to containerised deployment for production environments.

4 System Design and Implementation

This chapter details the design, architecture, and implementation of the fraud detection proof-of-concept system. The implementation follows the modular structure established during the design phase, with each component independently developed, tested, and integrated.

4.1 System Architecture

The system employs a modular, layered architecture comprising three primary components: a synthetic data generation module, a machine learning fraud detection engine, and an interactive Streamlit-based dashboard. All parameters are centralized in a JSON configuration file. The modular design ensures each component can be developed, tested, and modified independently, following established software engineering principles and supporting the iterative nature of the DSR methodology.

Table 4. System Architecture Components

Component	Technology Stack	Purpose
Data Generation	Python, Faker, NumPy, JSON config	Generate configurable synthetic transaction datasets
ML Detection Engine	scikit-learn, imbalanced-learn, pandas	Train Random Forest classifier, predict fraud
Interactive Dashboard	Streamlit, Matplotlib, Plotly	Real-time visualization, analysis, export
Configuration	JSON (config.json)	Centralized parameter management

The system is implemented as a set of modular Python files organized into three packages - data, ml, and ui - with a single entry point app.py that wires all components together. This structure is described in detail in Appendix 4.

4.2 Synthetic Data Generation Module

Since real banking transaction data could not be accessed due to privacy and regulatory constraints, the system generates synthetic data programmatically. Each transaction record contains: Transaction ID, User ID, transaction amount in local currency, a transaction type drawn from a four-category set covering online purchases, ATM withdrawals, bank transfers, and POS purchases, geographic location (country), timestamp, device type (Mobile, Laptop, Desktop, Tablet), currency determined by country-to-currency mapping, amount converted to USD using predefined exchange rates, and a binary fraud label (Is_Fraudulent).

Countries are not sampled uniformly but using a weighted probability distribution that reflects real-world transaction volume differences across regions. For example, the United States receives a weight of 10 compared to North Korea's weight of 1, making a US transaction approximately ten times more likely to appear in the dataset. All transaction amounts are converted to USD before applying fraud thresholds, ensuring fair cross-currency comparisons: a transaction of ₹50,000 Indian Rupees converts to approximately \$600 USD, correctly placing it below the \$4,000 high-value threshold rather than falsely flagging it as high-value based on the raw numeric amount.

The sidebar control shown in Figure 1 allows the user to configure the number of transactions to generate, with values ranging from 1,000 to 10,000, before clicking the Generate Dataset button to trigger the data generation module.

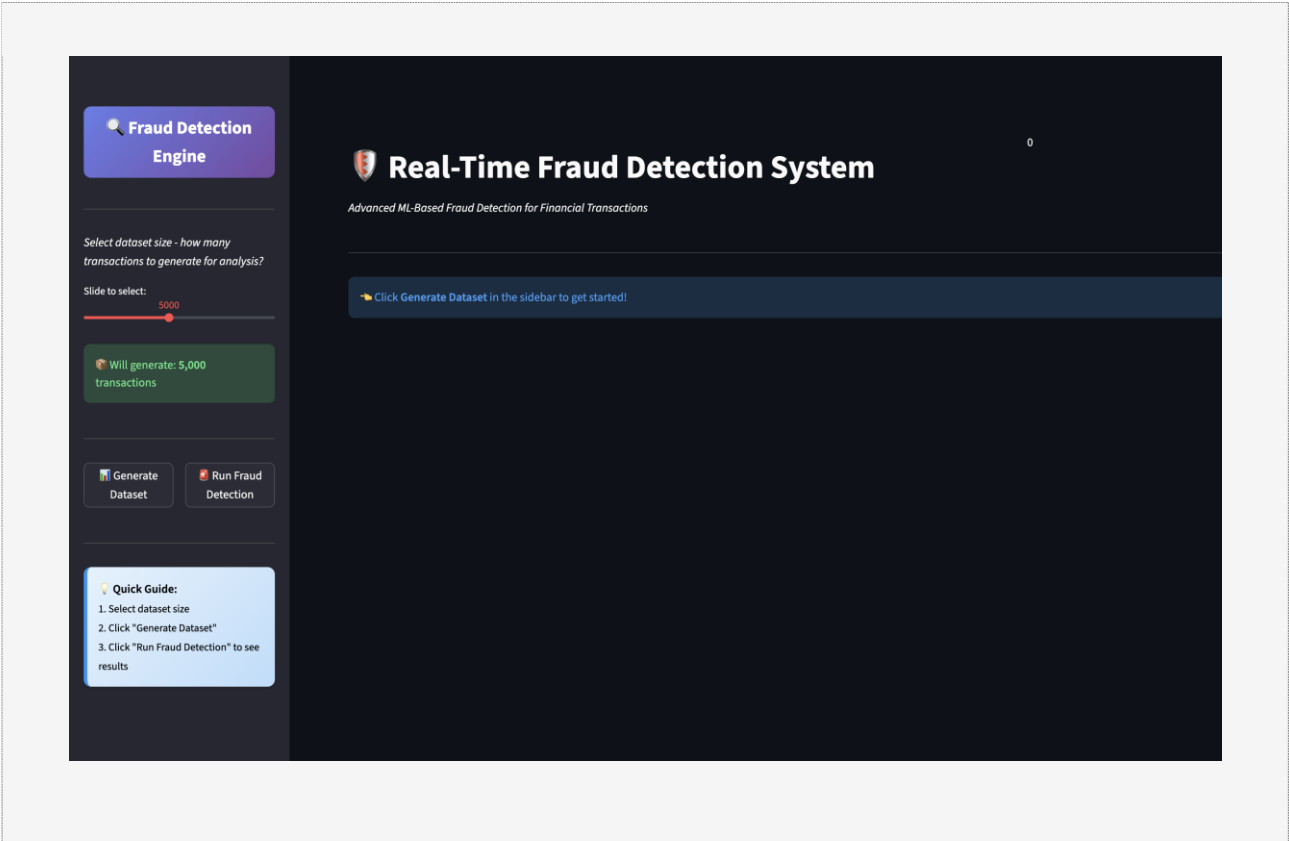


Figure 1. Configurable dataset generation controls in the sidebar

Once generation is complete, the system stores the DataFrame in Streamlit session state and presents a paginated preview table as shown in Figure 2. This table displays the key fields of the generated dataset, confirming that the data generation module has correctly produced transactions across multiple currencies, locations, and transaction types.

Transaction Data Preview

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Transaction_ID	User_ID	Amount	Amount_USD	Transaction_Type	Location	Timestamp	Is_Fraudulent
TXN1	USER9787	1746.68	1.92	Online Purchase	North Korea	2025-02-20 22:38:00	0
TXN2	USER6458	1221.7	14.66	Online Purchase	India	2025-02-20 15:48:00	0
TXN3	USER2530	1047.7	1.15	POS Purchase	Nigeria	2025-02-20 18:37:00	0
TXN4	USER3474	4415.82	48.57	Online Purchase	Russia	2025-02-20 05:10:00	0
TXN5	USER5747	4012.94	4333.98	Bank Transfer	Germany	2025-02-20 04:23:00	1
TXN6	USER8658	3111.11	34.22	POS Purchase	Russia	2025-02-20 08:21:00	0
TXN7	USER7486	1834.94	366.99	Bank Transfer	Brazil	2025-02-20 10:04:00	0
TXN8	USER4312	3730.28	746.06	Online Purchase	Brazil	2025-02-20 08:52:00	0
TXN9	USER4113	4739.77	31.76	Online Purchase	Japan	2025-02-20 12:41:00	0
TXN10	USER2291	4407.22	881.44	ATM Withdrawal	Brazil	2025-02-20 05:08:00	1

[Download Complete Dataset \(CSV\)](#)

Figure 2. Generated synthetic transaction dataset preview

Fraud labels are assigned probabilistically during generation based on three configurable risk conditions. These conditions are controlled entirely by parameters in config.json, meaning fraud rates and thresholds can be adjusted without modifying any Python code. The three conditions, read directly from the configuration file, are:

- **High-risk location:** if the transaction originates from Russia, Nigeria, North Korea, China, or Brazil, it receives a fraud label with probability 0.10 (10%).
- **Outside business hours:** if the transaction hour falls outside 06:00–22:00, it receives a fraud label with probability 0.30 (30%).
- **High-value amount:** if the Amount_USD exceeds \$4,000, it receives a fraud label with probability 0.40 (40%).

These conditions are independent and may co-occur. The following code extract from data/generator.py shows exactly how these probabilities are applied, demonstrating that the values are read from config rather than hardcoded:

```

def _assign_fraud(location, hour, amount_usd, config) -> int:
    if location in config["high_risk_locations"] and random.random() <
config["high_risk_prob"]:
        return 1
    if hour not in config["normal_hours"] and random.random() <
config["fraud_prob_outside_hours"]:
        return 1
    if amount_usd > config["high_amount_threshold"] and random.random() <
config["fraud_prob_large_amount"]:
        return 1
    return 0

```

Figure 3. Rule-based probabilistic fraud labeling logic

The call `random.random()` generates a random floating-point number between 0.0 and 1.0. The comparison `random.random() < 0.10` evaluates to True approximately 10% of the time, implementing the probabilistic fraud assignment. All threshold values (0.10, 0.30, 0.40, and \$4,000) are stored in `config.json` as shown in Table 5.

Table 5. Data Generation Configuration Parameters

Parameter	Default Value	Description
num_records	10,000	Number of transactions to generate
high_risk_locations	5 countries	Countries with elevated fraud probability (10%)
high_amount_threshold	\$4,000 USD	High-value transaction threshold
fraud_prob_outside_hours	0.30	Off-hours fraud probability
fraud_prob_large_amount	0.40	High-value fraud probability
high_risk_prob	0.10	High-risk location fraud probability

Parameter	Default Value	Description
normal_hours	06:00–22:00	Defined business hours

It is important to note that the conditions described above are used exclusively to assign ground-truth fraud labels during data generation and are not used by the detection model at inference time. The Random Forest classifier described in Section 4.4 receives only the engineered transaction features (amount, hour, is_night, location, transaction type, device) and the binary fraud label must independently learn which features are predictive of fraud. The thresholds in this section therefore serve as a research instrument for producing labelled synthetic data rather than as a component of the fraud detection system itself. In a production deployment, these synthetic labels would be replaced by real-world fraud confirmations such as chargebacks and customer disputes, with no change required to the detection pipeline.

4.3 Data Preprocessing and Feature Engineering

Before any data can be fed into a machine learning model, all columns must be converted to numbers. The raw dataset contains text columns and timestamp strings that must be transformed into meaningful numerical features through a process called feature engineering.

4.3.1 Temporal Feature Extraction

The Timestamp column stores dates and times as plain strings such as '2025-02-20 02:14:00'. The model cannot learn from a raw string, so two numerical features are extracted:

- **hour** - the hour of the day as an integer (0–23), extracted by slicing characters 11–13 of the timestamp string.
- **is_night** - a binary flag (1 or 0) set to 1 if the transaction occurred before 06:00 or after 22:00, directly encoding the time-based fraud signal identified in the data generation rules.

For example, a timestamp of '2025-02-20 02:14:00' produces hour = 2 and is_night = 1, while '2025-02-20 14:30:00' produces hour = 14 and is_night = 0. This feature extraction enables the model to learn time-based fraud patterns without requiring it to interpret raw datetime strings.

4.3.2 One-Hot Encoding of Categorical Variables

Transaction_Type, Location, and Device columns contain text categories. Machine learning models cannot process text directly and require all inputs to be numerical. There are two approaches to converting categories to numbers:

Label encoding - the wrong approach - assigns each category an integer (Mobile = 1, Laptop = 2, Desktop = 3, Tablet = 4). This is mathematically incorrect for nominal categories because it implies a false ordering: the model would interpret Tablet (4) as being four times greater than Mobile (1), or Desktop (3) as being numerically between Laptop and Tablet. No such relationship exists between device types.

One-hot encoding - the correct approach used in this system creates a separate binary column for each category. Each column answers a yes/no question: 'Was this transaction made on a Mobile device?' (1 = yes, 0 = no). This representation contains no false ordering: each device type is treated as entirely independent of the others. The parameter drop_first=True removes one reference column per category, as the dropped category is identifiable from all other columns being zero, preventing multicollinearity.

The same one-hot encoding is applied to Transaction_Type, Location, and Device columns. This results in the model receiving binary indicator columns such as Location_Nigeria, Transaction_Type_ATM-Withdrawal, Device_Mobile, and so on.

4.3.3 Class Imbalance Handling

The generated dataset contains approximately 83% normal transactions and 17% fraudulent ones. This class imbalance creates a serious problem: a naive model could achieve 83% accuracy simply by predicting 'Normal' for every transaction - without ever learning the fraud pattern. Two complementary techniques address this:

- **RandomOverSampler** - applied exclusively to the training set, this duplicates fraud rows until fraud and normal counts are equal (50% each). This forces

the model to give both classes equal attention during training. Crucially, the test set is never oversampled, as doing so would produce artificially optimistic evaluation metrics.

- **class_weight='balanced'** - this parameter instructs the Random Forest to penalise fraud misclassifications more heavily than normal misclassifications. With approximately 7,000 transactions in the training set (~83% normal, ~17% fraud), the effective fraud weight becomes approximately 5 times greater than the normal weight, pushing the model to prioritise catching fraud.

4.4 Random Forest Classifier

The choice of Random Forest was driven by four practical strengths reported in the fraud detection literature: resistance to overfitting on tabular data, native support for heterogeneous feature types, the capacity to model non-linear feature interactions, and built-in feature importance scoring useful for analyst review (Liu et al. 2015; Boyko & Mokryk 2021). A Random Forest is an ensemble of decision trees - individual flowcharts of yes/no questions about transaction features - that each independently vote on a prediction. The final prediction is determined by majority vote across all 100 trees, making the ensemble significantly more robust than any single tree.

Each tree is trained on a different random sample of the training data, and at each decision point considers only a random subset of features. This deliberate randomness ensures the trees are diverse and that their errors cancel out across the ensemble. The classifier is instantiated with the following parameters in ml/detector.py:

```
model = RandomForestClassifier(  
    n_estimators=100,      # build 100 decision trees  
    class_weight='balanced', # penalise fraud mistakes more heavily  
    random_state=42,      # fixed seed for reproducibility  
    n_jobs=-1,           # use all available CPU cores  
)
```

Figure 4. Random Forest classifier configuration

The parameter `random_state=42` ensures that the same random split is used every time the code runs, making results reproducible and comparable across experimental runs. The value 42 has no special significance - any fixed integer produces reproducible results.

The full workflow applied in this thesis is: split data 70% training / 30% test with stratified sampling to preserve the original fraud rate in both sets; apply `RandomOverSampler` to the training data only; train the classifier on the balanced training set; predict on the unseen test set; and evaluate performance using accuracy, precision, recall, F1-score, and confusion matrix analysis. Evaluating exclusively on the test set ensures that reported metrics reflect genuine generalisation to unseen transactions rather than memorisation of training patterns.

Beyond ML classification, the system provides explainable fraud reason labels assigned by a separate rule-based function. This function checks each flagged transaction's features against fraud thresholds in priority order: amount exceeding \$4,000 USD, location in the high-risk list, hour outside normal business hours, or a general suspicious activity label. This combination of ML prediction with rule-based explanation supports regulatory compliance with GDPR Article 22, which requires that individuals receive an explanation of automated decisions that significantly affect them.

4.5 Interactive Dashboard

The interactive visualization component is implemented as a Streamlit web application providing comprehensive fraud analysis through a modular user interface. The dashboard follows a sequential workflow: the user first generates a dataset, then runs fraud detection, and the results sections progressively reveal themselves as shown in the figures below.

4.5.1 Fraud Detection Results and Summary Metrics

After clicking Run Fraud Detection, the system trains the model and presents the results. The dashboard displays three key performance indicators: total fraud cases detected, the combined value of fraudulent transactions in USD, and the average fraud amount per transaction. Below these KPI cards, the flagged transactions table presents individual fraud cases sorted by `Amount_USD` in descending order, with the `Fraud_Reason` column providing a human-readable explanation for each flag.

Fraudulent Transaction Details (Showing 1,641 rows)

First Prev Next Last Page 2/165 | 10 rows/page

Transaction_ID	User_ID	Amount_USD	Location	Transaction_Type	Timestamp	Fraud_Reason
TXN7343	USER5696	5308.21	France	Bank Transfer	2025-02-20 07:21:00	High-value transaction flagged
TXN6405	USER6652	5301.63	France	ATM Withdrawal	2025-02-20 04:56:00	High-value transaction flagged
TXN31	USER8399	5291.32	France	Bank Transfer	2025-02-20 12:17:00	High-value transaction flagged
TXN3626	USER4212	5289.8	Germany	Online Purchase	2025-02-20 02:54:00	High-value transaction flagged
TXN9289	USER6678	5288.81	Germany	POS Purchase	2025-02-20 21:11:00	High-value transaction flagged
TXN7440	USER9251	5286.22	France	Bank Transfer	2025-02-20 03:39:00	High-value transaction flagged
TXN5044	USER7895	5285.8	Germany	Online Purchase	2025-02-20 02:32:00	High-value transaction flagged
TXN4841	USER6732	5283.48	France	ATM Withdrawal	2025-02-20 20:41:00	High-value transaction flagged
TXN7826	USER7566	5281.1	France	Online Purchase	2025-02-20 11:46:00	High-value transaction flagged
TXN8229	USER6007	5280.13	France	Bank Transfer	2025-02-20 01:20:00	High-value transaction flagged

Figure 5. Fraud detection results showing summary metrics and flagged transaction details

The Fraud_Reason column shown in Figure 5 provides one of four labels: 'High-value transaction flagged', 'Transaction from high-risk location', 'Unusual transaction time detected', or 'Suspicious activity detected'. It is important to note that this column is produced by a separate rule-based function and represents an explanation of why the transaction data looks suspicious it is not a direct window into the model's internal decision-making process.

4.5.2 Model Performance and Confusion Matrix

The model evaluation section of the dashboard presents accuracy, precision, recall, and F1-score alongside an interactive confusion matrix, as illustrated in Figure 6. The four gradient-coloured metric cards provide both the numerical value and an expandable explanation of what each metric means in the context of fraud detection, supporting analysts who may not have a machine learning background.

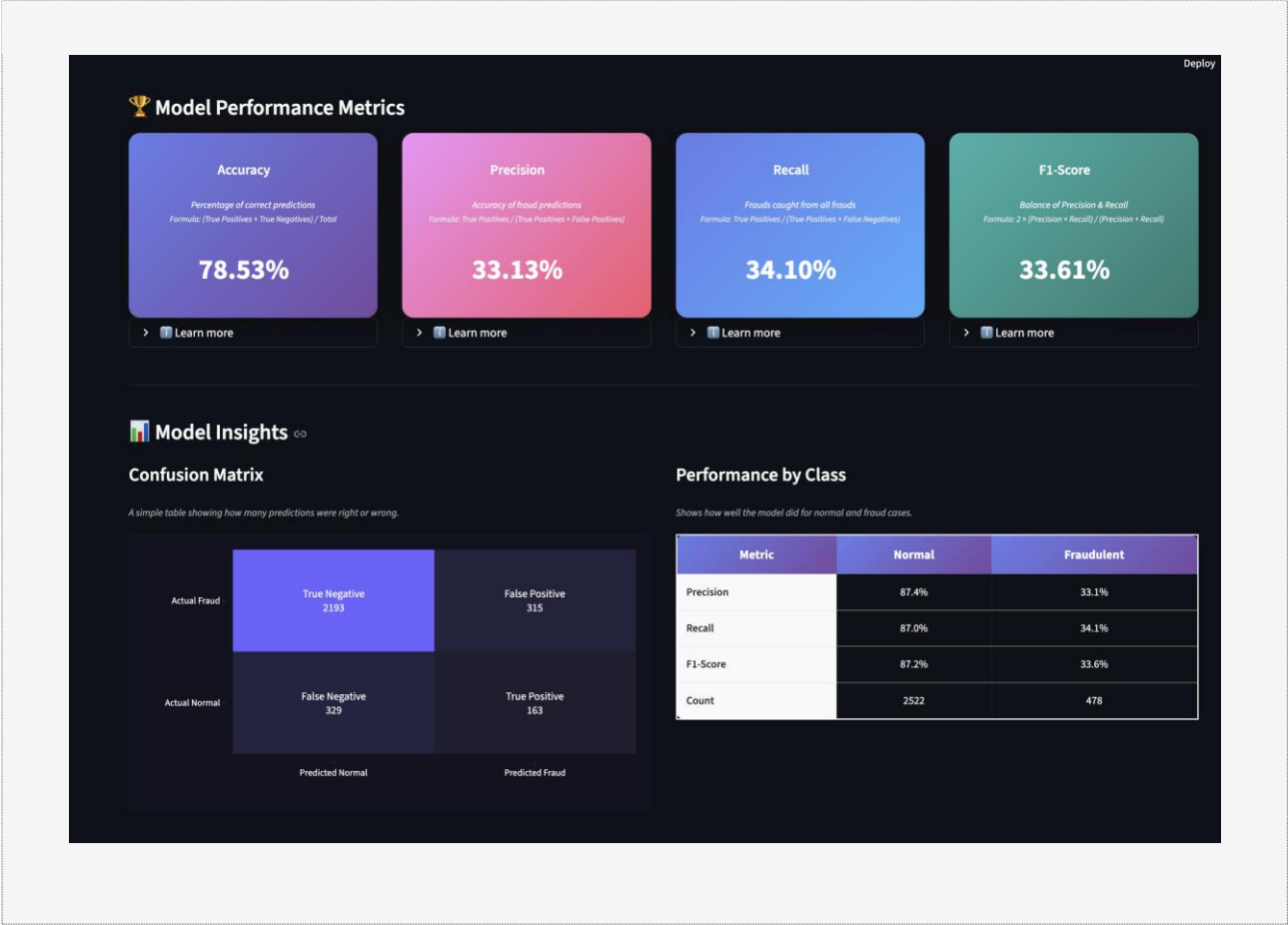


Figure 6. Model performance metrics and confusion matrix visualisation

The confusion matrix shown in Figure 6 breaks down prediction outcomes into four categories. True Negatives (TN) represent normal transactions correctly identified as normal. False Positives (FP) are normal transactions incorrectly flagged as fraud, generating unnecessary alerts. False Negatives (FN) are fraudulent transactions the model classified as normal - the most costly error type. True Positives (TP) are fraudulent transactions correctly detected. All four performance metrics are derived directly from these four counts, as detailed in Chapter 5.

4.5.3 Distribution Analysis

The distribution analysis section allows analysts to explore the dataset through multiple lenses: by transaction type, by location, and by amount range in USD. These charts use Matplotlib bar charts consistent with the original system's visual style. The section is available both for the full dataset (showing the overall distribution) and restricted to fraudulent transactions only (showing where fraud is concentrated). Full-page screenshots of these analysis views are provided in Appendix 5.

4.6 Configuration Management

All tunable system parameters are stored in a centralized JSON configuration file (config.json), enabling modification of system behaviour without requiring code changes. This configuration-driven approach supports systematic experimentation: different fraud scenarios can be tested by editing a single file and re-running the system, without touching any Python code. The full contents of config.json are provided in Appendix 1.

5 Results and Evaluation

The synthetic dataset consisted of 10,000 transactions distributed across 11 countries, 4 transaction types (Online Purchase, ATM Withdrawal, Bank Transfer, POS Purchase), and 4 device types (Mobile, Laptop, Desktop, Tablet). Fraudulent transactions constituted approximately 17% of the total dataset, reflecting a realistic albeit moderate class imbalance characteristic of financial fraud datasets. Transaction amounts ranged from small micro-transactions to high-value transfers exceeding \$10,000 USD after currency conversion.

Analysis of the generated dataset confirmed that fraud was disproportionately associated with the three configured risk factors: transactions originating from high-risk countries, high-value transactions exceeding the \$4,000 USD threshold, and transactions occurring outside normal business hours. This distribution validated the probability-based generation logic and demonstrated that the synthetic data module successfully produced datasets with realistic and interpretable fraud patterns suitable for ML training and evaluation.

5.1 Model Performance

The model was trained on the 70% training set after oversampling and evaluated exclusively on the 30% held-out test set. Table 6 reports the performance metrics achieved.

Table 6. Model Performance Metrics

Metric	Value	Interpretation
Overall Accuracy	78.53%	Proportion of all transactions correctly classified
Fraud Class Precision	33.13%	Of predicted fraud, proportion actually fraud
Fraud Class Recall	34.10%	Of actual fraud, proportion correctly identified
Fraud Class F1-Score	33.61%	Harmonic mean of fraud precision and recall

The overall accuracy of 78.53% indicates approximately four-fifths of transactions were correctly classified. The fraud-class metrics – precision of 33.13%, recall of 34.10%, and an

F1-score of 33.61% – reflect the inherent challenges of minority-class detection on a relatively small synthetic dataset. The near-equal values of precision and recall indicate a balanced error profile, meaning the model produces a similar number of false alarms and missed frauds rather than failing predominantly in one direction.

Even a modest detection rate has practical value. A system processing 100,000 transactions daily with a 0.5% fraud rate would detect approximately 170 fraudulent transactions per day at this recall rate, potentially preventing significant financial losses and providing a foundation for iterative system improvement.

5.2 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of prediction outcomes across the four possible combinations of actual and predicted labels. For a test set of 3,000 transactions (30% of 10,000), the four outcome categories are defined as follows:

- **True Negative (TN):** normal transactions correctly identified as normal. These require no action and represent the successful majority of predictions.
- **False Positive (FP):** normal transactions incorrectly flagged as fraud. These generate unnecessary investigations and customer disruption.
- **False Negative (FN):** fraudulent transactions incorrectly classified as normal. These represent undetected fraud and constitute the most costly error type.
- **True Positive (TP):** fraudulent transactions correctly identified. These represent successfully caught fraud cases.

Table 7. Confusion Matrix Summary

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	True Negatives (TN) - correctly identified normal transactions	False Positives (FP) - unnecessary fraud alerts
Actual Fraud	False Negatives (FN) - undetected fraud	True Positives (TP) - fraud correctly caught (~34.10%)

The current balance between false positives and false negatives is acceptable for a proof-of-concept but would require optimization for production deployment, where higher recall would be prioritized to minimize undetected fraud even at the cost of additional false alarms.

5.3 Feature Importance Analysis

The Random Forest classifier provides feature importance scores indicating which input variables most influenced the predictions. These scores represent the average reduction in impurity (Gini impurity) contributed by each feature across all 100 trees in the forest.

Table 8. Feature Importance Ranking

Rank	Feature	Relative Importance
1	Amount_USD (transaction amount)	Highest
2	High-risk country indicators	High
3	Transaction hour	Moderate-High
4	is_night indicator	Moderate

Rank	Feature	Relative Importance
5	Transaction type indicators	Lower
6	Device type indicators	Lowest

The most influential features align with the domain knowledge embedded in the data generation logic, confirming the model successfully learned the underlying fraud patterns. The prominence of transaction amount is consistent with real-world fraud detection findings (Du & Zhai 2024). The low feature importance of device type is expected and consistent with the data generation design, where device was not used as a fraud signal - the model correctly learned that device type alone carries minimal predictive value for fraud classification.

5.4 Impact of Dataset Size on Performance

The relatively modest fraud-class metrics can be substantially attributed to the limited number of fraud instances available for model training. With approximately 1,700 fraudulent transactions in the full dataset and only approximately 1,190 in the training set after the 70:30 split, the Random Forest has a constrained number of positive examples from which to learn discriminative patterns.

A second factor contributing to the modest fraud-class performance is the probabilistic nature of the fraud labelling itself. Because each of the three trigger conditions assigns a fraud label only 10%, 30%, or 40% of the time, the majority of transactions exhibiting a high-risk-location, off-hours, or high-value pattern are nevertheless labelled non-fraud during generation. This introduces label noise on the very features the model is being asked to learn from: identical or near-identical feature combinations can appear in both the fraud and non-fraud classes, so the model cannot draw a clean decision boundary on those features alone. The result is a fundamental ceiling on the achievable fraud-class precision and recall under the current generator configuration, independent of the dataset size or the algorithm chosen. A production deployment trained on real labels – derived from chargebacks, customer disputes, and confirmed fraud cases – would not face this synthetic-label noise and would therefore be expected to achieve substantially higher fraud-class metrics on a comparable feature set.

With larger datasets in the range of 50,000 to 100,000 transactions, the model would benefit from a substantially larger pool of fraud examples, reduced variance in model predictions due to more stable tree ensembles, more reliable decision boundaries at the class threshold, and consequently improved fraud-class precision, recall, and F1-score. The literature consistently supports the expectation that ML model performance in fraud detection improves significantly with increased data volume (Jahnavi et al. 2024; Johora et al. 2024). The current results successfully validate the proof-of-concept architecture and methodology; meaningful performance improvement with scale is both expected and achievable.

6 Discussion

This chapter discusses the findings in relation to the research questions, compares results with the existing literature, presents practical implications, and acknowledges limitations of the research.

6.1 Answering the Research Questions

RQ1: What are the key functional and non-functional requirements for developing an AI-powered fraud detection system in the banking sector?

The literature review and system implementation identified several key requirements. Functional requirements include the ability to process large volumes of transactions efficiently, automated fraud detection using adaptive machine learning models, near-real-time results delivery, transparent fraud reason labels for each flagged transaction, and configurable parameters for different fraud scenarios. Non-functional requirements include modularity for independent component updates, privacy compliance through synthetic data usage, usability through an intuitive dashboard interface, and reproducibility through documented processes and configuration-driven architecture.

RQ2: How can machine learning algorithms be effectively utilized to detect fraudulent activities in banking transactions?

The Random Forest algorithm proved effective when combined with appropriate preprocessing techniques. One-hot encoding preserved the non-ordinal nature of categorical features. Temporal feature extraction enabled time-based fraud pattern learning. Currency normalization standardized amounts across currencies for meaningful comparisons. RandomOverSampler addressed class imbalance without contaminating the evaluation set. The model achieved 78.53% overall accuracy, and feature importance analysis confirmed successful pattern learning with transaction amount, geographic risk indicators, and timing as the most influential features.

RQ3: What are the major challenges and limitations in implementing an AI-based fraud detection system, and how can they be addressed?

Several challenges were encountered and addressed. Class imbalance was handled through Random Oversampling and balanced class weights. Data unavailability was mitigated through configurable synthetic data generation. The interpretability challenge was addressed through fraud reason labels providing transparent justifications alongside predictions. Limited

fraud-class performance (precision and recall around 33-34%) was attributed to the relatively small dataset of 10,000 transactions, consistent with literature indicating ML models require substantially larger datasets for robust minority-class performance.

RQ4: How does the proof-of-concept demonstrate the feasibility and effectiveness of AI technologies in enhancing fraud detection capabilities compared to traditional methods?

The proof-of-concept successfully demonstrates AI feasibility through a complete end-to-end pipeline. Compared to traditional rule-based methods, the ML approach offers adaptive pattern learning from data, simultaneous multi-factor analysis, probabilistic risk scoring beyond binary flagging, and quantified feature importance providing data-driven insights. The interactive dashboard enables analysts to explore results through multiple perspectives including geographic analysis, temporal patterns, amount distributions, and performance metrics.

6.2 Comparison with Literature

The findings of this thesis are broadly consistent with the existing literature on machine learning-based fraud detection. Across multiple comparative studies, Random Forest emerges as a strong performer on tabular fraud datasets, with the literature citing its overfitting resistance, ability to mix numerical and categorical features, and interpretable importance rankings as key advantages (Liu et al. 2015; Boyko & Mokryk 2021; Chang et al. 2022; Du & Zhai 2024). The overall accuracy of 78.53% achieved here is within the expected range for a proof-of-concept system operating on a relatively small synthetic dataset.

Class imbalance and the value of oversampling are well-documented topics in the fraud detection literature. Chawla et al. (2002) established the foundational understanding of the class-imbalance problem, and a more recent review by Moreira et al. (2022) reinforces the conclusion that oversamplers - whether random duplication or synthetic generation through SMOTE - improve minority-class metrics provided that the oversampling step is restricted to the training partition. The feature-importance findings of this thesis, in which transaction amount and geographic location emerge as the dominant predictors, mirror the patterns observed independently by Du and Zhai (2024) and Patil et al. (2024). This convergence between the synthetic-data feature importances and real-world findings supports the use of the generator as a credible stand-in when actual banking data is unavailable.

6.3 Practical Implications

The synthetic data generation module provides a reusable and extensible tool for researchers and practitioners who require realistic transaction data for experimentation without access to proprietary banking records. By making fraud indicators configurable through a JSON configuration file, the module enables systematic experimentation with different fraud scenarios, risk thresholds, and dataset compositions.

The modular system architecture demonstrates how a fraud detection pipeline can be structured for maintainability and extensibility in practice. Each component can be independently updated, tested, and replaced without affecting the other modules, particularly relevant for financial institutions that need to evolve their detection capabilities incrementally while maintaining system stability.

The fraud reason labelling feature addresses the growing practical and regulatory need for explainable AI decisions in financial services, providing human-readable justifications that support both operational decision-making and regulatory compliance with frameworks such as GDPR. This dual approach of combining ML predictions with rule-based explanations provides a template for systems that must satisfy both predictive accuracy and regulatory transparency requirements.

6.4 Limitations

Several limitations of this thesis should be acknowledged. First, the use of synthetic data, while necessary due to privacy constraints, may not fully capture the complexity, noise, and subtlety of real-world banking fraud patterns. Real transaction data contains nuanced customer behaviour patterns, seasonal trends, and adversarial adaptation that are difficult to replicate synthetically.

Second, the single-algorithm approach using only Random Forest limits the generalizability of performance conclusions; a multi-algorithm comparison would provide more robust evidence of algorithmic suitability. Third, the modest fraud-class metrics, ranging from 33.13% to 34.10% for precision and recall, with an F1-score of 33.61%, highlight the constraints imposed by the relatively small dataset and the label noise inherent in the probabilistic data-generation process. While these metrics validate the proof-of-concept, they would be insufficient for production deployment where higher detection rates are essential.

Fourth, the system currently lacks sequential transaction analysis that would capture patterns across multiple transactions by the same user. Fifth, the batch processing approach does not support true real-time streaming, which would be necessary for production deployment where transactions must be scored within milliseconds. These limitations collectively define clear and actionable directions for future enhancement.

7 Conclusion

This thesis successfully designed, implemented, and evaluated a proof-of-concept system for AI-powered fraud detection in banking, achieving all stated research objectives through a structured design science research methodology. The system integrates three primary modules forming a comprehensive fraud detection pipeline: a configurable synthetic data generation module that produces realistic transaction datasets with controllable fraud indicators; a Random Forest-based machine learning engine employing one-hot encoding, temporal feature extraction, currency normalization, and RandomOverSampler for class imbalance handling; and an interactive Streamlit dashboard providing real-time visualization, analysis, and export of fraud detection results.

The system achieved 78.53% overall accuracy on the held-out test set and demonstrated the ability to learn meaningful fraud patterns from the synthetic data, as confirmed by feature importance analysis showing transaction amount, geographic risk indicators, and transaction timing as the most influential predictors. The implementation of transparent fraud reason labels alongside ML predictions addresses the growing regulatory and operational demand for explainable AI in financial services. All four research questions were systematically addressed through the combined evidence of the literature review, system implementation, and quantitative evaluation results.

7.1 Key Contributions

This thesis makes several contributions to both the academic and practitioner communities. First, it provides a comprehensive end-to-end proof-of-concept integrating synthetic data generation, machine learning classification, and interactive visualization into a single, cohesive system, demonstrating the complete fraud detection pipeline rather than focusing on isolated algorithmic components.

Second, the configurable synthetic data generation module offers a reusable and extensible tool addressing the persistent challenge of data unavailability in fraud detection research. By enabling researchers to control fraud indicators, dataset size, and risk parameters through a JSON configuration file, the module supports systematic experimentation without requiring access to sensitive banking data.

Third, the emphasis on transparency through fraud reason labelling represents a practical contribution to the growing field of explainable AI in financial services. The dual approach of

combining ML predictions with rule-based explanations provides a template for systems that must satisfy both predictive accuracy and regulatory transparency requirements.

7.2 Recommendations for Future Work

- Real-time streaming integration using Apache Kafka or Apache Flink for continuous transaction monitoring within milliseconds of transaction initiation.
- Live currency rate APIs for dynamic, real-time currency conversion replacing the static exchange rates used in this proof of concept.
- Multi-algorithm comparison incorporating XGBoost, LightGBM, neural networks, and ensemble voting to provide more comprehensive evidence of algorithmic suitability.
- Increased dataset scale (50,000–100,000+ transactions) to provide substantially more fraud examples for training and improve fraud-class recall and precision.
- User behaviour profiling incorporating sequential transaction analysis over time to capture patterns such as rapid successive withdrawals or unusual geographic movement.
- Cloud deployment on AWS, GCP, or Azure with Docker containerization and Kubernetes orchestration for production-scale systems.
- Sophisticated fraud scenarios including coordinated fraud rings, gradual amount escalation patterns, and adversarial adaptation to detection rules.

In conclusion, this thesis demonstrates that machine learning-based fraud detection is a viable and promising approach for the banking sector. The proof-of-concept system provides a solid, modular, and extensible foundation upon which future development can build toward production-ready fraud detection capabilities. The combination of configurable data generation, supervised machine learning with transparent explanations, and interactive visualization creates a comprehensive toolkit that serves both as a practical demonstration and as a platform for continued research and development.

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Appendices

Appendix 1. System Configuration (config.json)

The following JSON configuration file controls all tunable parameters of the fraud detection system. Fraud thresholds, dataset size, high-risk locations, and transaction types can all be modified here without changing any Python code.

```
{
  "num_records": 10000,
  "high_risk_locations": ["Russia", "Nigeria", "North Korea", "China", "Brazil"],
  "country_weights": {
    "United States":10, "India":9, "China":9, "Brazil":7, "Nigeria":6,
    "Germany":5, "Russia":5, "France":4, "Japan":3, "Australia":2, "North Korea":1
  },
  "fraud_prob_outside_hours": 0.3,
  "fraud_prob_large_amount": 0.4,
  "high_amount_threshold": 4000,
  "high_risk_prob": 0.1,
  "normal_hours": [6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22],
  "transaction_types": ["Online Purchase", "ATM Withdrawal", "Bank Transfer", "POS Purchase"],
  "device_types": ["Mobile", "Laptop", "Desktop", "Tablet"]
}
```

Appendix 2. Country-Currency Mapping

Country	Currency Code	USD Conversion Rate
United States	USD	1.000
India	INR	0.012
China	CNY	0.140
Brazil	BRL	0.200
Nigeria	NGN	0.0011
Germany	EUR	1.080
Russia	RUB	0.011
France	EUR	1.080
Japan	JPY	0.0067
Australia	AUD	0.650
North Korea	KPW	0.0011

Appendix 3. Sample Fraudulent Transactions

Transaction_ID	Amount_USD	Location	Type	Hour	Fraud_Reason
TXN4521	\$4,820	Russia	Bank Transfer	02:00	High-value transaction flagged
TXN7834	\$3,100	Nigeria	ATM Withdrawal	23:00	Unusual transaction time detected
TXN1203	\$4,200	China	Online Purchase	14:00	High-value transaction flagged
TXN9901	\$850	North Korea	POS Purchase	09:00	Transaction from high-risk location

Appendix 4. Project File Structure

The following structure shows the modular organization of the fraud detection system implementation.

```
fraud_app/
├─ app.py                (Entry point – wires all components together)
├─ config.json           (All configurable parameters)
├─ requirements.txt      (Python dependencies: pandas, numpy, faker,
│                         streamlit, scikit-learn, imbalanced-learn,
│                         plotly, matplotlib)
├─ data/
│   ├─ __init__.py
│   └─ generator.py      (Synthetic data generation)
├─ ml/
│   ├─ __init__.py
│   └─ detector.py       (ML pipeline: encoding, training, evaluation)
└─ ui/
    ├─ __init__.py
    ├─ styles.py         (Global CSS styling)
    ├─ sidebar.py        (Sidebar controls – slider and buttons)
    ├─ overview.py       (KPI metrics and paginated data preview)
    ├─ charts.py         (Matplotlib chart functions)
    ├─ results.py        (Fraud table, confusion matrix, metric cards)
    └─ export.py         (CSV download buttons)
```

Appendix 5. Fraud Analysis Distribution Charts

The following figure presents the fraud analysis section of the dashboard, showing the distribution of flagged transactions by fraud reason and by geographic location.

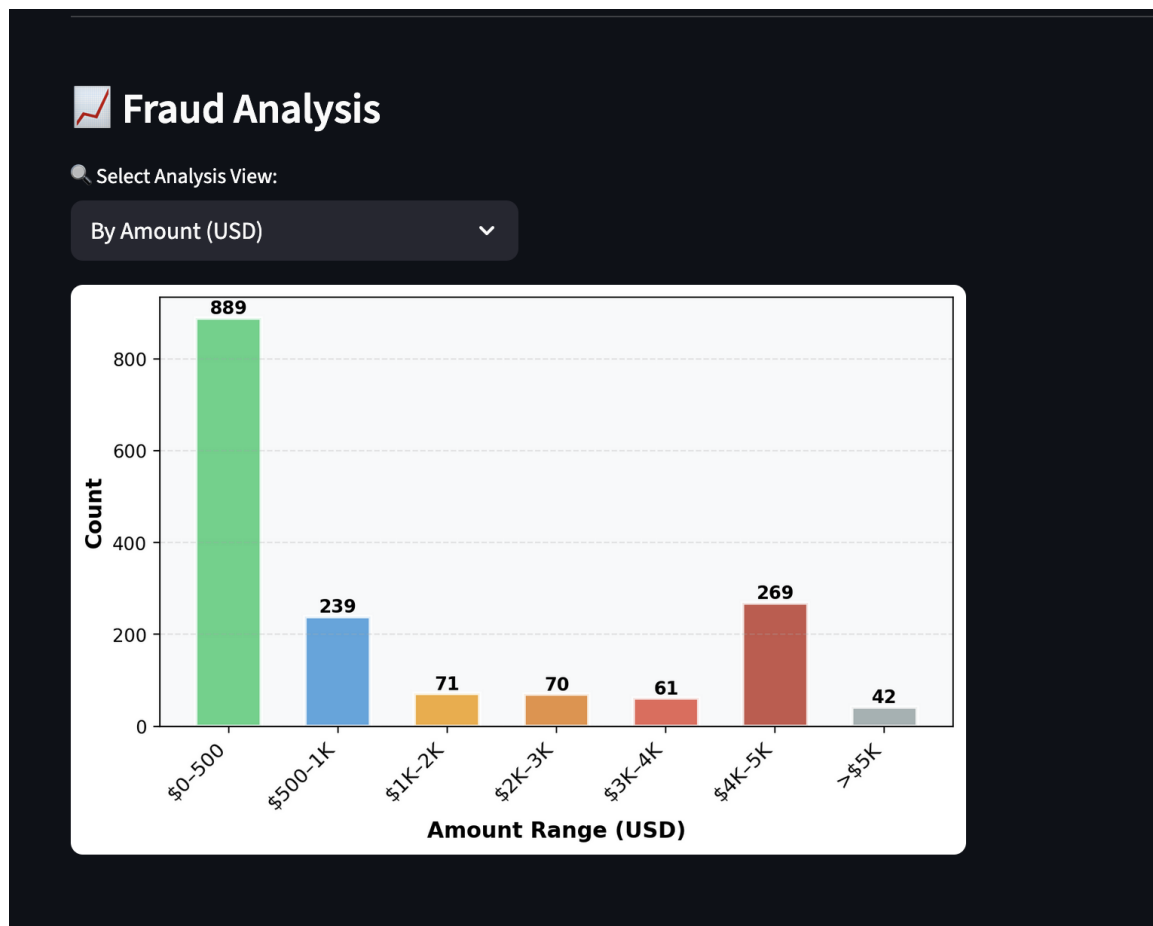


Figure A: Fraud Analysis by Transaction Amount Range

Fraud Analysis

Select Analysis View:

By Location

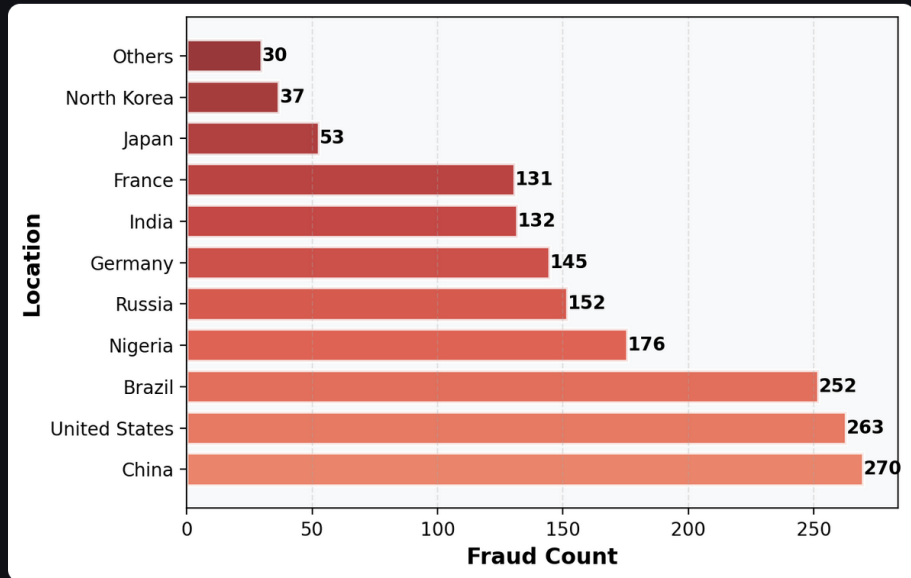


Figure B: Fraud Analysis by Geographic Location

Fraud Analysis

Select Analysis View:

By Transaction Type

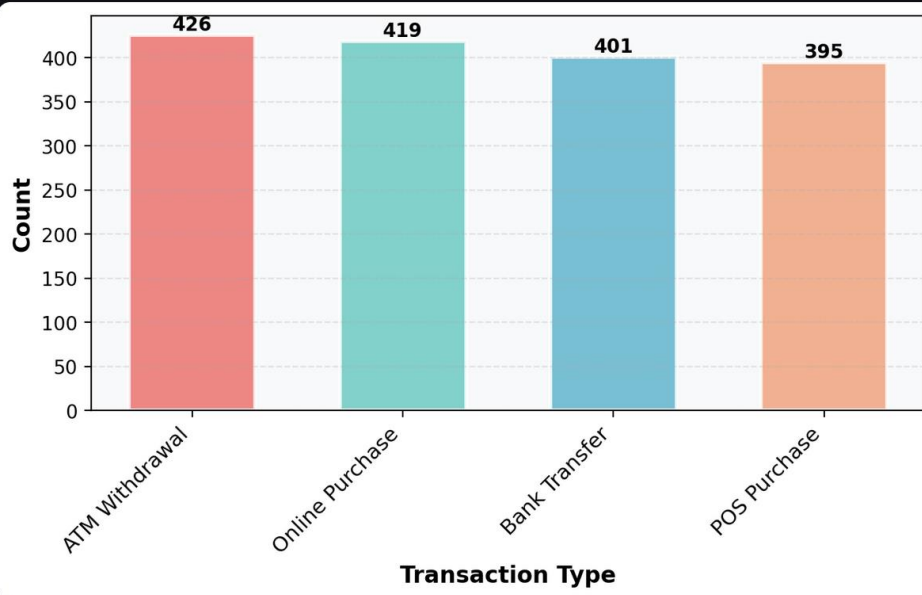


Figure C: Fraud Analysis by Transaction Type

Fraud Analysis

Select Analysis View:

By Fraud Reason

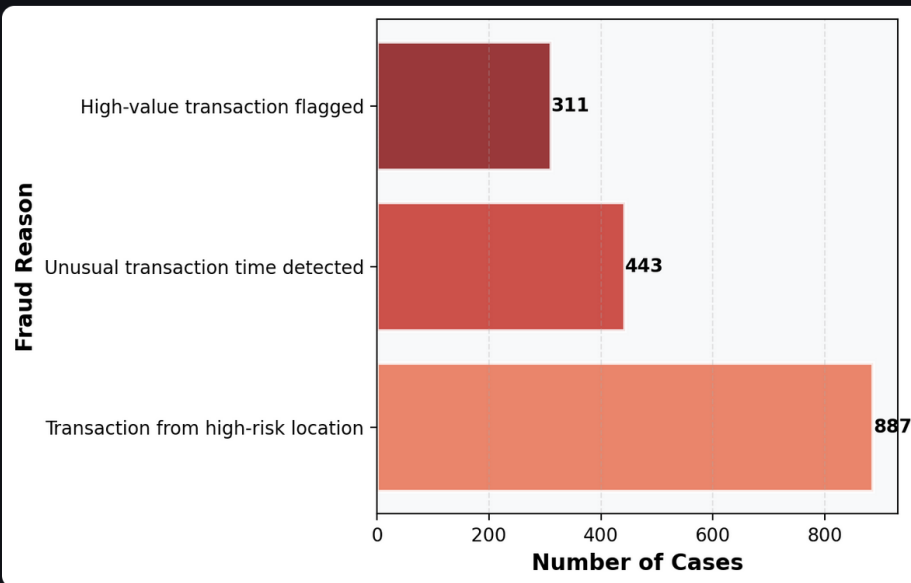


Figure D: Fraud Analysis by Detected Fraud Reason